

Sentinel-Conditioned 24-Hour-Ahead Irradiance Forecasting over Distributed Photovoltaic Fleets

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Abstract. Accurate 24-hour-ahead solar irradiance forecasting over spatially distributed photovoltaic (PV) fleets is important for renewable energy dispatch, grid operation, and market participation. However, forecasting over large groups of PV plants remains challenging because high-quality local measurements are often available only at a limited number of monitored sites, while irradiance fields exhibit strong spatial and temporal variability.

This paper addresses PV irradiance forecasting as a spatio-temporal learning problem with sparse observations. A limited set of monitored PV sites, referred to as sentinel sites, is used to inform predictions of future irradiance trajectories across the entire fleet. The proposed sentinel-conditioned framework combines spatially joint field modelling, attention-based conditioning from the sentinel sites, and temporal encoding of the previous 24 hours of observations. PV module temperature is included as a separate auxiliary input channel, while the forecasting target remains irradiance-only.

Evaluated over a full validation year (8,760 hours), the proposed framework outperforms persistence across the 24-hour forecast horizon, achieving nRMSE of **11.3%** and KGE of **0.89**, with the largest relative improvements at longer lead times. The results indicate that sparse local measurements can improve one-day-ahead irradiance forecasting when combined with spatially joint learning over a PV fleet.

Keywords: Photovoltaic generation · Solar irradiance forecasting · One-day-ahead forecasting · Machine learning · Sparse sensing · Spatio-temporal forecasting

1 Introduction

The increasing penetration of photovoltaic (PV) generation is making accurate solar irradiance forecasting a key component of grid operation, energy trading, and renewable energy dispatch [3]. In particular, 24-hour-ahead forecasting is relevant for operational planning, market participation, and managing variability in power systems with large shares of renewable generation.

In practice, however, forecasting performance is often limited by the availability and quality of local measurements. Satellite-derived products and Numerical Weather Prediction (NWP) models provide broad spatial coverage, but they

may not fully capture local microclimatic effects or plant-specific environmental dynamics. Conversely, high-quality on-site observations can provide more accurate local information, but deploying and maintaining instrumentation at every photovoltaic (PV) plant becomes costly and impractical as the number of monitored sites increases.

This trade-off motivates forecasting approaches that exploit a limited number of high-fidelity local measurements while preserving accurate predictions over an entire PV fleet. In this work, we investigate whether a small set of strategically selected PV sites can be used to condition 24-hour-ahead irradiance forecasts across both monitored and non-monitored locations. We refer to these instrumented sites as *sentinel sites*. Unlike classical sensor placement problems, candidate sensor locations cannot be freely optimized over a continuous spatial domain, but are constrained to the finite set of sites composing the PV fleet. The resulting problem can therefore be interpreted as a discrete, sparse-sensing counterpart of physical field reconstruction, adapted to the photovoltaic forecasting setting. Throughout this paper, a *PV fleet* denotes a set of geographically distributed PV sites jointly modeled as a spatio-temporal irradiance field. The term *site* refers to the elementary spatial unit of the analysis, rather than an individual PV module or panel.

Recent sparse-observation field reconstruction methods provide a useful methodological starting point for this problem. In particular, PhySense [8] is a flow-matching framework designed to reconstruct physical fields from a limited number of optimally placed sensors.

This approach integrates sensor-conditioned attention with field-based neural modelling, enabling the propagation of information from sparse measurements across irregular spatial domains. In this work, we apply this concept to a different context: instead of reconstructing a static physical field from instantaneous observations, we aim to predict the future irradiance field over 24 hours for a fleet of PV sites. This requires incorporating temporal histories, restricting the candidate sentinel locations to existing PV sites, and conditioning the forecast on a fixed set of monitored sentinel sites.

The resulting contribution is a sentinel-conditioned sparse observation framework for 24-hour-ahead irradiance forecasting over a spatially distributed PV fleet. The framework combines three main ideas. First, the selection problem is formulated over discrete existing PV sites, identifying which sites should be monitored and used as sentinels. Second, temporal histories of irradiance and, optionally, PV module temperatures are encoded to capture recent diurnal and meteorological dynamics. Third, sparse sentinel observations are injected into a flow-matching forecasting backbone to produce spatially joint irradiance trajectories for all PV sites in the fleet.

2 Related Work

Single-site photovoltaic forecasting. A large part of the PV forecasting literature addresses irradiance or power prediction at the level of individual sites [11, 1].

These methods typically rely on historical irradiance, historical PV power, or meteorological inputs, and use statistical, recurrent, or sequence-based learning architectures to capture diurnal and seasonal temporal patterns [4]. While effective for isolated point forecasts, single-site approaches treat each PV plant independently and do not explicitly exploit the spatial correlations that arise among neighboring or regionally distributed sites. When applied to a large PV fleet, this reduces the spatial field to a collection of independent temporal sequences and discards information that may be useful for spatially joint forecasting.

Multi-site spatio-temporal forecasting. The limitations of independent-site modelling have motivated a second line of research on multi-site spatio-temporal forecasting, where PV plants are modelled jointly, and information from one location can support prediction at other locations. Simeunović et al. [14] proposed graph-convolutional Long-Short Term Memory (LSTM) and transformer models trained on production data from up to 1,000 PV systems, showing that graph-based spatial aggregation can improve forecast accuracy at horizons up to six hours. Subsequent methods introduce self-adaptive adjacency matrices to capture dynamic spatial correlations across sites [18]. More recent approaches, such as SolarFM [13], exploit time-series foundation models and spatio-temporal encoders to combine historical observations with forecasted sequences. These works demonstrate the value of spatially joint learning, but they generally assume that the relevant input information is available at all or most target sites, rather than explicitly addressing the sparse-observation setting considered here.

Auxiliary meteorological features. Photovoltaic power production is mainly governed by the solar irradiance incident on the modules, typically expressed as global irradiance on the array plane. In standard PV performance models, the produced power is primarily related to plane-of-array irradiance, while temperature affects the conversion efficiency through thermal losses and temperature-dependent performance coefficients [2]. Accordingly, irradiance is the dominant variable in PV forecasting, whereas air temperature or PV module temperature provides a physically meaningful secondary source of information. When direct module-temperature measurements are not available, module temperature is often estimated from ambient temperature, irradiance, and, in some models, wind speed [15].

Data-driven PV forecasting models may use meteorological variables in different ways. Some approaches rely only on historical irradiance or power measurements, whereas operational forecasting models may also include future exogenous inputs from NWP models or weather data services [17]. However, variables such as humidity, pressure, wind speed, cloud cover, or aerosol indicators should not be concatenated indiscriminately, since weakly informative or noisy features can increase dimensionality and may even degrade forecast generalisation [12, 9].

In this work, we consider irradiance as the primary variable, retained both as PV site representation and variable to predict. Then, we perform some experiments.

Sparse-observation field reconstruction and flow matching. Beyond forecasting-specific architectures, recent work on sparse-observation physical field reconstruction is relevant to the present setting because it addresses the problem of inferring a spatial field from a limited number of measurements. Generative and physics-informed approaches have demonstrated good spatial fidelity on irregular geometries compared with graph-based or convolutional baselines [10, 6].

Among these methods, PhySense [8] is particularly relevant because it combines flow matching, sensor placement, and attention-based conditioning to reconstruct physical fields from sparse measurements. Its original formulation, however, is designed for static field reconstruction and assumes a physical domain in which sensor locations can be optimised. The PV forecasting setting considered here differs in three main respects: candidate sentinel locations are restricted to existing PV sites, the target is a future 24-hour irradiance trajectory rather than a contemporaneous field measurement, and temporal histories must be encoded alongside spatial information. These differences motivate the adaptation developed in the proposed sentinel-conditioned forecasting framework.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{P} = \{\mathbf{x}_i \in \mathbb{R}^2\}_{i=1}^N$ denote the fixed spatial coordinates of the N PV sites considered in this study. At each physical time step t , site i is associated with a vector of input variables $\mathbf{z}_i^t \in \mathbb{R}^C$, where C denotes the number of available channels. The collection $\{\mathbf{z}_i^t\}_{i=1}^N$ is interpreted as a multivariate spatio-temporal field defined over the PV fleet. Observations are assumed to be available at hourly resolution.

Given an input horizon of length T_{in} , the historical sequence at site i is

$$\mathbf{h}_i = (\mathbf{z}_i^{t-T_{\text{in}}+1}, \dots, \mathbf{z}_i^t) \in \mathbb{R}^{T_{\text{in}} \times C}.$$

The forecasting target is the future irradiance trajectory over a horizon of length T_{out}

$$\mathbf{y}_i = (g_i^{t+1}, \dots, g_i^{t+T_{\text{out}}}) \in \mathbb{R}^{T_{\text{out}}},$$

where g_i^t denotes irradiance at site i and time t . Thus, even when multiple input channels are available, the target considered in this paper is irradiance-only; auxiliary variables, such as PV module temperature, are used only as conditioning information.

The joint forecasting problem is therefore to learn a spatial forecasting operator

$$F_\theta : \{\mathbf{x}_i, \mathbf{h}_i\}_{i=1}^N \mapsto \{\hat{\mathbf{y}}_i\}_{i=1}^N, \quad (1)$$

conditioned, in the sparse-observation setting, on the observations available at a subset of monitored sentinel PV sites.

We identify a single-channel task: irradiance forecasting based only on historical irradiance. Each site features a time series, which is then $\mathbf{z}_i^t = g_i^t \in \mathbb{R}_{\geq 0}$.

In the dual-channel case, the input vector contains irradiance and PV module temperature, $\mathbf{z}_i^t = (g_i^t, \theta_i^t) \in \mathbb{R}^2$, where θ_i^t denotes PV module temperature. The corresponding history contains the previous T_{in} hourly values of both channels, while the forecasting target remains the irradiance trajectory only. Throughout the paper, $T_{\text{in}} = T_{\text{out}} = 24$ hours, corresponding to 24-hour-ahead forecasting from the preceding 24-hour observation window.

3.2 Sentinel selection

In large portfolios of spatially distributed PV plants, collecting real-time or high-quality measurements from every site may be impractical because of instrumentation, communication, and maintenance costs. Therefore, this work considers a sparse-observation setting in which only a reduced subset of PV sites is monitored. These monitored sites act as sentinels, providing local observations that condition the forecast over the entire PV fleet.

Let

$$\mathcal{S} \subset \{1, \dots, N\}$$

denote the index set of monitored sentinel PV sites. The forecasting problem in Eq. 1 then becomes

$$F_\theta : \{\mathbf{x}_s, \mathbf{h}_s\}_{s \in \mathcal{S}} \mapsto \{\hat{\mathbf{y}}_i\}_{i=1}^N, \quad (2)$$

where forecasts for the entire PV fleet are generated from observations collected only at the sites belonging to $\mathcal{S}$.

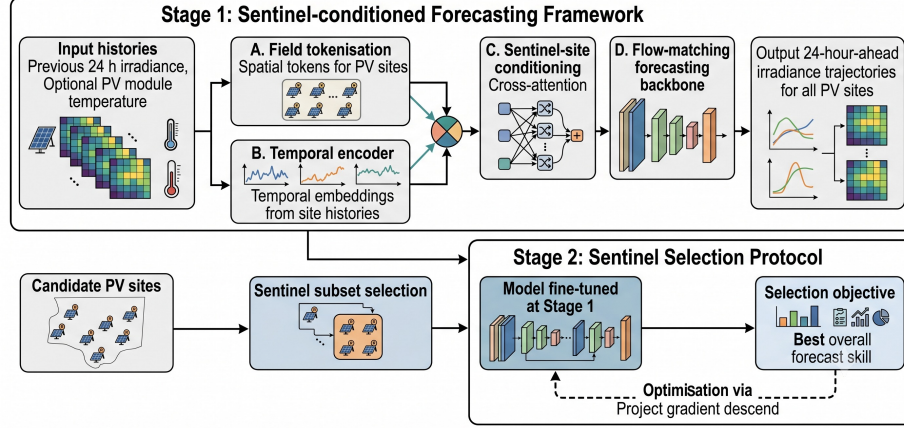
Beyond forecasting, the objective is to identify the subset of monitored PV sites that maximises the predictive information available to the model. This is methodologically related to the sensor placement optimisation framework introduced in PhySense [8], where sparse sensors are selected or optimised to reconstruct a physical field. In the present setting, however, the candidate locations are not arbitrary points in a continuous domain, nor are they sensors placed within a single PV plant; rather, they are existing PV sites within a distributed PV fleet. The optimal sentinel set is defined as

$$\mathcal{S}^* = \arg \min_{\mathcal{S}} \mathcal{L} \left(\{\mathbf{y}_i\}_{i=1}^N, F_\theta(\{\mathbf{x}_s, \mathbf{h}_s\}_{s \in \mathcal{S}}) \right), \quad (3)$$

where $\mathcal{L}$ denotes the forecasting loss evaluated over the entire PV fleet. The PV sites belonging to $\mathcal{S}^*$ are referred to as the sentinels of the PV fleet.

The setting considered here differs from the original PhySense formulation because the candidate locations are discrete existing PV sites within a spatially distributed fleet. Sentinel selection is therefore interpreted as the selection of the PV sites to be monitored, rather than as the placement of sensors within a single photovoltaic plant. The selected sentinel sites provide a sparse but informative sampling of the irradiance field, while the forecasting model learns to propagate their information through the spatial correlations observed across the PV fleet. As a result, the objective extends the continuous physical field reconstruction with an additional focus on forecasting the future irradiance state of all PV sites conditioned on a fixed set of monitored sentinel sites.

3.3 Architecture of the sentinel-conditioned forecasting framework



The proposed architecture builds on flow-matching field modelling and sparse conditioning [8], but it is designed for a forecasting setting. In the PV application considered here, the model receives temporal histories from a limited set of monitored sentinel sites and predicts the future irradiance trajectories over the full PV fleet. This requires combining temporal encoding, spatial information propagation, and sentinel-conditioned attention within a 24-hour-ahead forecasting framework.

The overall workflow is summarised in Fig. 3.3 and is explained step by step in the following paragraphs. The main blocks are: Step A, field tokenisation; Step B, temporal encoding; Step C, sentinel-site conditioning; and Step D, the flow-matching forecasting backbone.

Step A – Field tokenisation.

The irradiance forecast is modelled as a spatial field defined over the PV fleet. To make this field usable by Transolver, each PV site is represented by a site-level token comprising its geographic coordinates, the current flow-matching state associated with the future irradiance trajectory, and geometric features derived from distances to a fixed reference grid. Transolver [16] avoids the quadratic attention cost of standard transformers by projecting the input tokens into a set of learned latent representations, known as slices. Self-attention is performed within this reduced slice space rather than directly between all site pairs, decreasing the computational complexity from $O(N^2)$ to $O(N \times M)$, where N denotes the number of sites and M the number of slices. The processed slice representations are subsequently decoded back to the site level, allowing efficient modelling of spatial interactions over large PV fleets.

For each site i , the spatial token is built as

$$\mathbf{r}_i = [\mathbf{x}_i; \tilde{\mathbf{y}}_i^{(\tau_{\text{fm}})}; \mathbf{d}_i] \in \mathbb{R}^{2+T_{\text{out}}+16}, \quad (4)$$

where $\mathbf{x}_i$ denotes the site coordinates, $\mathbf{d}_i$ is a 16-dimensional vector of distances to a fixed 4×4 reference grid, and $\tilde{\mathbf{y}}_i^{(\tau_{\text{fm}})}$ denotes the intermediate flow-matching state. The artificial time τ_{fm} should not be confused with the physical time index t of the irradiance series. A shared linear pre-processor maps each token to the hidden dimension d , producing the spatial representation processed by the forecasting backbone.

Step B – Temporal encoding. Forecasting requires conditioning on the recent temporal evolution of each PV site. The temporal encoder summarises the previous 24 hours of observations into a latent temporal embedding. For each site, the history $\mathbf{h}_i \in \mathbb{R}^{T_{\text{in}} \times C}$ is flattened into a single vector and processed by a two-layer fully connected neural network with GELU activation:

$$\text{TemporalEncoder}(\mathbf{h}_i) = W_2 \sigma(W_1 \text{vec}(\mathbf{h}_i) + b_1) + b_2 \in \mathbb{R}^d, \quad (5)$$

where σ is the activation function. This embedding captures recent diurnal evolution, variability, and meteorological dynamics, using irradiance alone in the single-channel case, and irradiance plus PV module temperature in the dual-channel case.

The temporal embedding is injected into the corresponding spatial field token, so that each site-level representation contains both where the PV site is and what its recent temporal behaviour has been. This mechanism is central to the adaptation from static sparse-field reconstruction to spatio-temporal 24-hour-ahead forecasting.

Step C – Sentinel-site conditioning. A limited subset of monitored PV sites provides high-quality local observations. These sentinel sites condition the forecast over the full PV fleet. For each sentinel site s , the model encodes its spatial coordinates and recent observation history into a sentinel token, for example

$$\mathbf{z}_s = [\mathbf{x}_s; \text{vec}(\mathbf{h}_s)].$$

These tokens are injected into the forecasting backbone via cross-attention, enabling each target PV site to leverage information from the most relevant monitored sites. In this way, sparse local measurements are propagated across the spatial field and used to guide the 24-hour-ahead irradiance forecast.

During fine-tuning, features at non-sentinel sites may be estimated from the selected sentinel sites by Inverse Distance Weighting (IDW):

$$\hat{\mathbf{z}}_i = \frac{\sum_{s \in \mathcal{S}} w_{is} \mathbf{z}_s}{\sum_{s \in \mathcal{S}} w_{is}}, \quad w_{is} = \|\mathbf{x}_i - \mathbf{x}_s\|_2^{-2}. \quad (6)$$

Step D – Flow-matching forecasting backbone. The final forecasting block generates future irradiance trajectories for the entire PV fleet. Starting from a noisy or intermediate representation of the future irradiance field, the model

learns a velocity field that progressively transforms this state into a plausible 24-hour-ahead forecast. The generation process is conditioned on spatial site tokens, temporal embeddings, and information from the sentinel sites.

During training, the intermediate state is not obtained from a previous forecast step. It is synthetically constructed by interpolating between Gaussian noise ϵ_i and the true future irradiance trajectory $\mathbf{y}_i$:

$$\tilde{\mathbf{y}}_i^{(\tau_{\text{fm}})} = \tau_{\text{fm}}\mathbf{y}_i + (1 - \tau_{\text{fm}})\epsilon_i.$$

When $\tau_{\text{fm}} = 0$, the state is pure noise; when $\tau_{\text{fm}} = 1$, it coincides with the true future trajectory. The model predicts the velocity field required to move from the noisy state toward the true target direction.

Following the conditional flow-matching formulation of [7], the model is trained with the objective

$$\mathcal{L} = \mathbb{E}_{\tau_{\text{fm}}, \epsilon} \left\| F_{\theta} \left(\tilde{\mathbf{y}}^{(\tau_{\text{fm}})}, \tau_{\text{fm}} \right) - (\mathbf{y} - \epsilon) \right\|_2^2. \quad (7)$$

During the forecasting phase, the future target is unknown. The model, therefore, starts from noise and updates the state using Euler integration steps until a 24-hour irradiance trajectory is obtained. In the experiments, $K = 5$ Euler steps are used.

3.4 Training Protocol

Training proceeds in two stages, following the distinction between generic sparse conditioning and the final fixed sentinel-site configuration.

Stage 1 – Full-field pre-training. The model is trained on all N PV sites, with sets of monitored sentinel-site indices sampled uniformly at random ($|S| \sim \text{Uniform}(10, 200)$) at each step. This exposes the model to diverse conditioning configurations and encourages generalisation to different subsets of monitored PV sites.

Stage 2 – Sentinel-site fine-tuning. The model is fine-tuned with observations fixed to the selected monitored sentinel PV sites. Features at non-sentinel PV sites are constructed via Equation (6). This stage aligns the model with the actual deployment configuration and improves forecast accuracy for the fixed sentinel-site layout.

All experiments use the Adam optimiser with a learning rate of 10^{-4} and a cosine annealing schedule over 300 epochs. The model has $d = 374$ hidden dimensions, 12 Transolver layers, and $d_{\text{cond}} = 93$. Flow-matching integration uses $K=5$ Euler steps during the forecasting phase. Training is performed on a single NVIDIA RTX PRO 6000 BLACKWELL GPU.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate the proposed framework over a regular spatial grid covering the Piedmont region, with a resolution of 5×5 km. The grid comprises $N = 1,149$

PV-site candidate locations, each identified by fixed geographic coordinates $\{\mathbf{x}_i\}_{i=1}^N$. Each coordinate represents a possible location for a PV installation, for which the model forecasts the corresponding solar irradiance profile.

Hourly irradiance and surface temperature data are available for all site locations through the Photovoltaic Geographical Information System (PVGIS) [5], providing a complete spatio-temporal description of the deployment. The dataset spans 17 years (2005–2022) for training and one year (2023) for validation, totaling 8,760 consecutive hourly observations.

While PVGIS data are available across the entire deployment and are used for model training and evaluation, they represent satellite- and model-derived estimates rather than direct on-site measurements. Our objective is to emulate a realistic operational setting in which higher-fidelity local sensors can only be deployed at a limited number of locations. In this context, sentinel plants represent instrumented sites where accurate in-situ observations are assumed to be available, while forecasts must still be produced for the entire deployment. The complete PVGIS field is therefore used for supervision during training, whereas sentinel conditioning is employed to examine how a sparse set of local measurements can improve forecasting performance across all sites by propagating spatial information.

Normalisation. We set $T_{\text{in}} = T_{\text{out}} = 24$ hours throughout. All irradiance values are normalised to $[0, 1]$ using the training-set minimum and maximum; temperature is independently normalised to $[0, 1]$. The physical denominator $G_{\text{max}} = 1,000 \text{ W/m}^2$ (Standard Test Conditions peak irradiance [3]) is used for all normalised evaluation metrics.

Evaluation protocol. Validation is performed over the test-set sliding hourly forecast windows, obtained by excluding the first and last $T_{\text{in}}=24$ hours of the validation year to ensure complete input and output sequences. For each window starting at timestep t , the model receives the irradiance (or irradiance and temperature) histories $\{\mathbf{h}_i\}_{i=1}^N$ over hours $[t-24, t)$ and produces a joint forecast $\{\hat{\mathbf{y}}_i\}_{i=1}^N$ over hours $[t, t+24)$. Errors are then computed between the forecast and ground truth over all N sites and all $T_{\text{out}}=24$ horizons. This protocol yields a per-timestep error curve of exactly $W=8,713$ points, one per forecast issue time across the validation year.

We compare our models against the following baselines. Each model is trained using the same protocol for data splitting and the learning task.

- **SolarFM** [13]: A foundation-model-driven solar forecasting architecture that combines Chronos-based probabilistic predictions, dual-channel ConvLSTM encoders, and tensor regression to capture both temporal and spatial dependencies in multi-site irradiance forecasting.

To make the algorithm fit our data, we scaled from eight stations to 1,149 spatial input locations. Then, the forecasting task shifted from a four-step prediction to direct 24-hour forecasting. Finally, instead of nine meteorological and irradiance variables, the adapted model used irradiance-derived inputs

and Chronos-generated future representations based on PVGIS data. The historical context window remained at 24 hours, in line with the original SolarFM design.

- **GRU**: The Gated Recurrent Unit (GRU) is considered to be a recurrent model baseline that takes the data from PV sites as independent time series to predict the same number of independent time series, T_{out} , snapshots in advance. We used 2 GRU layers with a hidden dimension of 128.
- **GCN-GRU**: The GCN-GRU baseline extends the previous model by correlating each PV site information via a message passing Graph Convolutional Network (GCN) on the neighbors. The adjacency matrix connecting the nodes is dynamically computed at each layer via a self-attention module on the node features. The spatial embeddings were subsequently processed by the GRU layer. We used 2 GCN and 2 GRU layers, each with a hidden dimension of 64.

Metrics. We report the normalised Mean Bias Error (nMBE), Mean Absolute Error (nMAE), and Root Mean Squared Error (nRMSE), together with Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (SMAPE), Mean Directional Accuracy (MDA), Nash–Sutcliffe Efficiency (NSE), Kling–Gupta Efficiency (KGE), and Mean Absolute Scaled Error (MASE). Together, these metrics provide a complementary assessment of forecasting performance by quantifying systematic bias, average prediction error magnitude, sensitivity to large deviations, relative percentage errors, directional consistency, and overall agreement between observed and predicted irradiance values. MBE measures the tendency of a model to overestimate or underestimate irradiance, while MAE and RMSE evaluate the magnitude of forecast errors, with RMSE assigning greater weight to larger errors. MAPE and SMAPE express errors in relative terms, facilitating comparisons across different irradiance levels. MDA assesses a model’s ability to correctly capture the direction of change between consecutive observations. NSE and KGE provide broader measures of predictive skill by evaluating how closely forecasts reproduce the variability and temporal dynamics of the observed irradiance series. Finally, MASE expresses forecast errors relative to a simple baseline forecast, enabling scale-independent comparisons across models.

Normalised metrics (nRMSE, nMAE, nMBE) are obtained dividing by $G_{\text{max}} = 1,000 \text{ W/m}^2$, corresponding to the Standard Test Conditions peak irradiance. MAPE and SMAPE are computed on daytime windows only (mean irradiance $> 1 \text{ W/m}^2$) to avoid instability caused by division by near-zero values during nighttime periods, while for nMBE we report the absolute value.

4.2 Results

We first evaluate the spatial forecasting performance of the proposed PhySense-based model against the other competing approaches. Figure 1 shows the geographical distribution of the nMAE across all $N = 1,149$ PV sites in the

validation set. Each point corresponds to a PV site location, colored by the average forecasting error across all prediction windows.

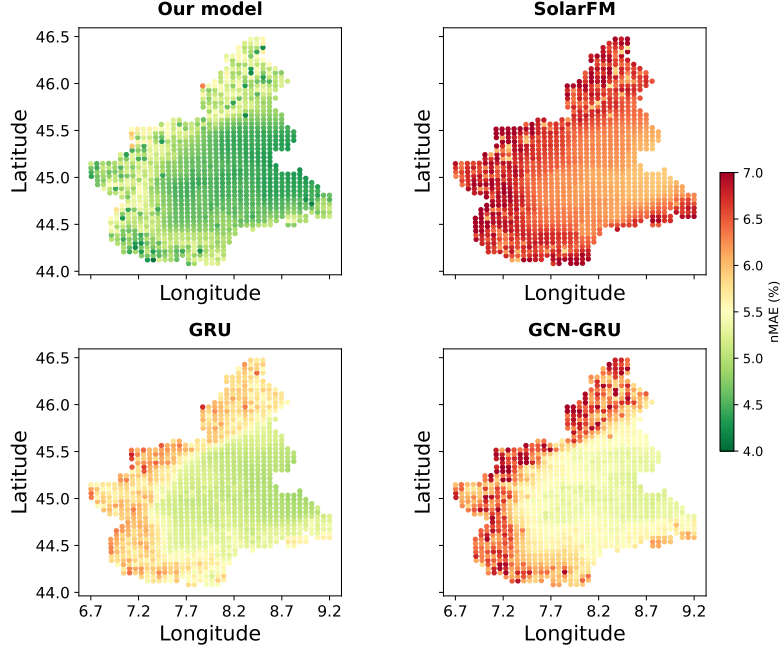


Fig. 1. Spatial distribution of forecasting error. Each marker represents a PV site location in the considered spatial grid. Markers are coloured according to the nMAE (%). Lower values indicate more accurate forecasts.

The spatial error map shows a clear geographic structure, with the lowest per-site nMAE observed in the central and eastern lowland areas and higher errors concentrated along the western and northern mountainous boundaries. This indicates that the model captures the irradiance field most accurately in spatially homogeneous regions, while performance degrades in topographically complex zones where local variability is stronger and harder to reconstruct from the fixed sentinel set.

To quantify the forecasting performance of the proposed model relative to the baselines, we report the aggregate results in Table 1. The table summarises validation performance over $W = 8,713$ forecast windows and $N = 1,149$ PV plants. Among the reported metrics, nMAE provides the most direct measure of average forecast accuracy relative to the irradiance range, SMAPE captures the

magnitude of relative error symmetrically, and KGE reflects the combined effects of correlation, bias, and agreement in variability.

Model	nMBE(↓) [%]	nMAE(↓) [%]	nRMSE(↓) [%]	MAPE(↓) [%]	SMAPE(↓) [%]	MDA(↑) [%]	NSE(↑) [-]	KGE(↑) [-]	MASE(↓) [-]
Our Model	0.02	4.81	11.3	61	56	41.3	0.795	0.890	2.484
SolarFM	1.35	6.49	10.8	88	125	26.1	0.813	0.869	3.351
GRU	0.25	5.49	10.2	91	116	46.6	0.833	0.863	0.970
GCN-GRU	0.52	5.85	10.5	101	119	25.3	0.823	0.844	3.023

Table 1. Validation results over $W=8,713$ forecast windows on $N=1,149$ sites. Upward and downward arrows denote metrics that should be maximized (↑) or minimized (↓), respectively. Best result per column in **bold**.

4.3 Forecast degradation over the 24-hour horizon

While the aggregate metrics in Table 1 provide a single summary of overall forecast quality, they do not reveal how quickly each model deteriorates as the forecast horizon increases. To better understand temporal robustness, Fig. 2 reports the horizon-wise evolution of nMAE, nRMSE, and KGE over the full 24-hour prediction window.

Our model exhibits the most stable behaviour across the horizon. In terms of nMAE, it increases more gradually than the competing methods and remains consistently below SolarFM, GRU, and GCN-GRU at nearly all lead times. The nRMSE curves show a different behaviour from nMAE. Although our model does not attain the lowest nRMSE across the full horizon, its curve is notably smoother and more stable than those of the competing models. This indicates that the model produces fewer large deviations and a more uniform error profile over time, even if its average squared error remains slightly higher than the strongest baseline in this metric.

The KGE curves provide an additional view of model robustness. Our model maintains the highest KGE values across most of the forecasting window, indicating stronger agreement in correlation, bias, and variability structure. In contrast, the baseline models show a more pronounced decline, particularly SolarFM, whose KGE decreases steadily as the horizon grows. Overall, the figure confirms that our model is not only accurate on average but also more resilient to horizon expansion than the competing methods.

4.4 Sensitivity to the candidate sensors’ choice

To assess how sensitive the second-stage optimisation is to the number of candidate sensors, we performed an ablation study in which the available sensor pool was progressively reduced to 10%, 30%, 50%, 70%, and 90% of the full set.

For each setting, the second-stage sentinel optimisation was rerun on the reduced pool and reported in, and the resulting models were evaluated on the same validation protocol. Figure 3 reports the results for 4 different site pools. Each pool

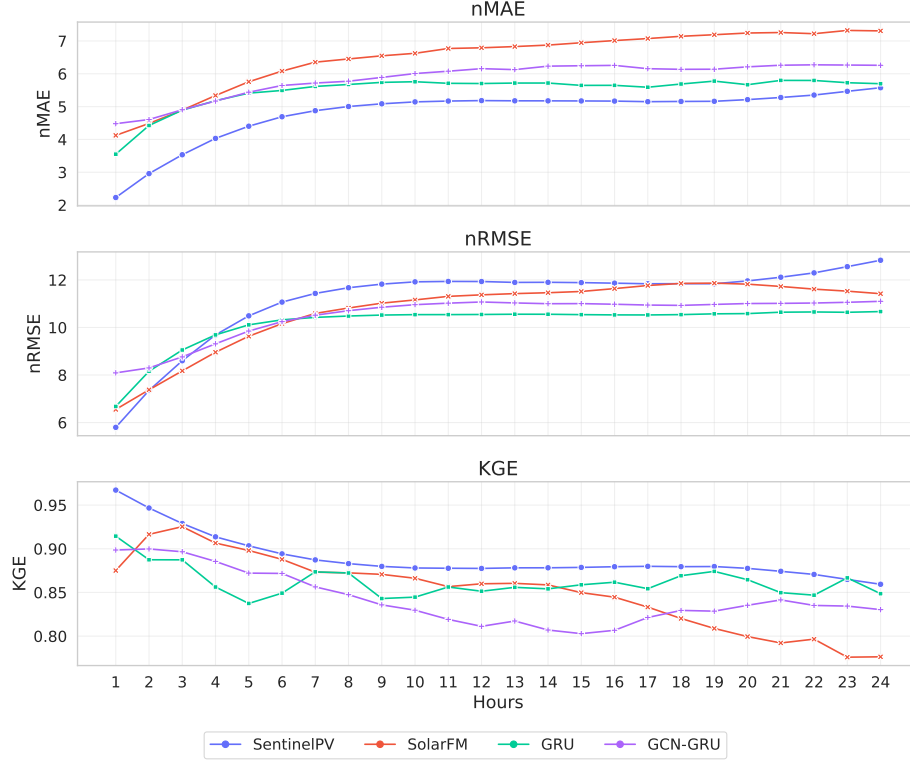


Fig. 2. Horizon-wise forecasting degradation over 24 hours of our model against the baselines. Lower is better for nMAE and nRMSE, while higher is better for KGE.

has been obtained by ordering the available sites by latitude or longitude, either in decreasing or increasing order, and sampling the first available. Therefore, each run requires using a portion of the sites in one of the four borders to predict the entire PV fleet. Therefore, this experiment isolates the effect of sensor availability on the quality of the learned configuration.

For each setting, the second-stage sentinel optimisation was rerun on the reduced pool of sites, and the results were reported accordingly, evaluating each model using the same validation protocol. Figure 3 presents the results for four different site pools. Each pool was created by ordering the available sites based on latitude or longitude, either increasing or decreasing, and then sampling the first available sites. Consequently, each run uses a portion of the sites located at one of the four borders to predict the entire PV fleet. This experiment, therefore, isolates the effect of sensor availability on the quality of the learned configuration.

The nMAE curve shows that performance is fairly stable across the tested ranges, with only small variations between settings. The best nMAE is obtained at 90% availability, while 30% also performs competitively and slightly outperforms

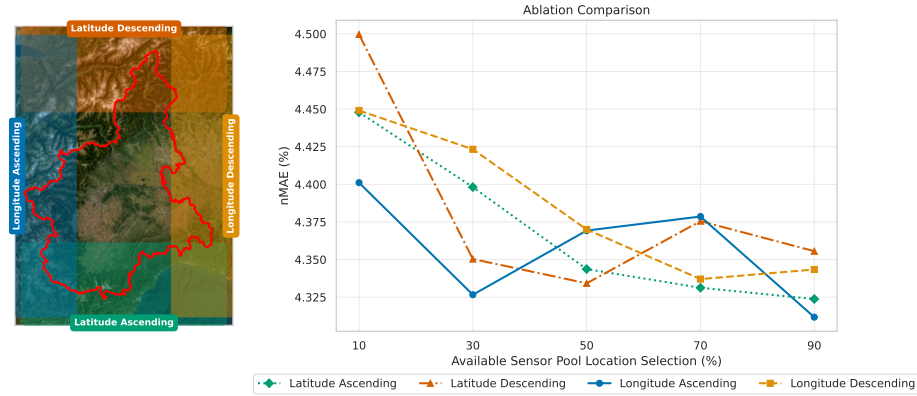


Fig. 3. nMAE across reduced pools of available sensors. As represented by the boxes on the map at the left of the figure, each pool selects the available locations by ascending latitude (corresponding to the bottom of the map), descending latitude (corresponding to the top), ascending longitude (corresponding to the left), and descending longitude (corresponding to the right).

70%. The overall spread is narrow, suggesting that the optimisation procedure is robust and can still identify effective sensor layouts even when the candidate pool is substantially reduced. In practical terms, this means the method does not require the full set of sensors to achieve near-optimal accuracy.

Overall, the ablation suggests that the second-stage sentinel optimisation is robust to substantial reductions in the available sensor pool. While the largest pool yields the best or near-best scores, the improvements beyond 30% are marginal, implying that the learned sensor placement strategy is not overly dependent on exhaustive candidate coverage. This is important for deployment scenarios where only a subset of sensors may be available for optimisation. It is important to note that the generalizability observed in one location may be due to the close proximity of each point within the PV fleet in our experiment. We should expect this property to degrade as we move to a national-level analysis.

5 Conclusion

We presented a sentinel-conditioned framework for 24-hour-ahead irradiance forecasting over distributed PV fleets. The main idea is to use a limited number of monitored sites, called sentinel sites, to condition forecasts for the entire fleet, including non-monitored locations. The proposed method combines spatial field modelling, temporal encoding of the previous 24 hours, sentinel-site attention, and a flow-matching forecasting backbone.

Experiments on 1,149 sites covering the entire Piedmont (Northern West of Italy) over a 5x5 km grid show that the proposed model improves average

forecasting accuracy compared with SolarFM, GRU, and GCN-GRU baselines. It achieves the best nMAE, SMAPE, KGE, and bias values, and shows more stable behaviour across the 24-hour forecast horizon. The spatial analysis indicates that the model performs best in homogeneous lowland areas, while errors increase in mountainous regions with stronger local variability.

The sensitivity study indicates that effective sentinel configurations can be identified even with a reduced sensor pool, making the approach applicable in practical monitoring scenarios with limited sites. Future work should validate the method on real in-situ measurements, extend it to larger and more diverse PV fleets, and incorporate additional meteorological or topographic information to improve performance in complex regions.

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